

SPINE EXPERT INDIA

Patient Digital Intake Workflow — Final Specification | SEI-Intake v4.0

This document is the final, build-ready specification for patient data collection across the SEI platform. It replaces the earlier single long questionnaire with three separate layers, so patients answer each question exactly once, and only when it is clinically relevant.



THE THREE LAYERS

- | | |
|----|---|
| 1. | Registration (one-time, before/at first appointment) — permanent profile + consents. Generates a Unique Patient ID . Never asked again unless the patient edits their profile. |
| 2. | Clinical Pre-Consult Questions (every appointment) — symptoms only. Triggered fresh each time an appointment is confirmed, generating a new Consult ID . Capped at 24 scored questions, with a small conditional add-on for higher-risk neck cases. |
| 3. | Document Upload (every appointment, optional per visit) — reports and imaging, auto-tagged and version-controlled. |

Once all three are complete, the system confirms the appointment slot and sends a teleconsultation link — see the final page of this document for the exact confirmation flow.

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Module 1 — Registration (One-Time Only)

Completed once, before or at the first appointment. Stored permanently in the patient's profile and auto-loaded into every future consult — never asked again unless the patient chooses to edit it.

Basic Details

- ☐ Full Name, Date of Birth, Age (auto-calculated), Gender
- ☐ Mobile Number, Email, City, Country, Preferred Language for consultation
- ☐ Emergency Contact — Name, Number, Relationship

Body Profile

- ☐ Height (cm), Weight (kg) — **BMI auto-calculated**

Occupation & Lifestyle

- ☐ Usual work: Desk/computer · Standing or walking · Manual/heavy labour · Homemaker · Student · Retired
- ☐ Activity level: Sedentary · Moderate · Heavy
- ☐ Smoking: Current · Past · Never | Alcohol: Yes · No

Medical History (multi-select)

- ☐ Diabetes · Hypertension · Heart disease · Kidney disease · Thyroid disease · Rheumatoid/autoimmune disease · Osteoporosis · Cancer · Parkinsonism · Tuberculosis · Other · None
- ☐ Current medicines — maintained as an editable list (patient taps **"Same as before"** or **"Update"** at each visit, instead of re-entering)
- ☐ Drug allergies — maintained as an editable list

Previous Surgical History

- ☐ Previous spine surgery — if yes: Year, Hospital, Level, Procedure
- ☐ Other surgeries: Joint replacement · Cardiac · Brain · Abdominal · Other

CONSENTS — completed once at registration

- ☐ **Consent 1 — Clinical Care (Mandatory):** "I consent to Spine Expert India collecting and using my information for clinical consultation and treatment."
- ☐ **Consent 2 — Privacy & Data Protection (Mandatory):** "I understand my information will remain confidential and stored securely as per applicable healthcare regulations."
- ☐ **Consent 3 — Research Registry (Optional):** "I voluntarily agree that my anonymised data may be included in the SEI National Outcomes Registry for research and quality improvement."
- ☐ **Digital signature + Date + OTP verification**

REGISTRATION COMPLETE → UNIQUE PATIENT ID GENERATED

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Module 2 — Clinical Pre-Consult Questions (Every Appointment)

Triggered fresh each time an appointment is confirmed. The system auto-loads the patient's registration profile in the background and issues a new **Consult ID** — the patient only answers what has changed since their last visit: today's symptoms.

NEW CONSULT ID GENERATED ON EVERY APPOINTMENT

SECTION A — URGENT RED-FLAG PRE-SCREEN (answered first, one screen)

Any positive answer (A–F) triggers an immediate urgent alert to the clinical team.

- ☐ A. Loss of bladder or bowel control
- ☐ B. Rapidly increasing weakness in arm or leg (hours to days)
- ☐ C. Fever with spine pain
- ☐ D. New/severe spine pain after a fall, accident, or major trauma
- ☐ E. Known cancer with new or worsening spine pain
- ☐ F. Significant unexplained weight loss in the past 3 months
- ☐ G. None of the above

Question budget for this module: 18 universal core questions + 6 adaptive tail questions = **24 scored questions**, capped. One additional outcome-tracking item (self-rated spine health) is counted separately, as it feeds longitudinal registry tracking rather than same-visit diagnosis. A small conditional add-on (2 questions) appears only for higher-risk neck cases — see Section C.

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Module 2 — Section B: Universal Core Questions (Q1–Q18)

Asked of every patient, one question per screen, with a progress bar.

#	Question	Response Options	Purpose
Q1	Primary reason for consultation	Neck pain · Back pain · Arm pain · Leg pain/sciatica · Numbness · Weakness · Walking difficulty · Follow-up · Second opinion	Routing
Q2	Where is your pain?	Interactive body-map tap	Routing
Q3	Which side?	Right · Left · Both · Midline	Clinical
Q4	How long have you had this problem?	<2 weeks · 2 weeks–3 months · 3–6 months · >6 months	Clinical
Q5	How did it start?	Gradual · Lifting · Accident · Posture · Exercise · After surgery	Clinical
Q6	Pain level right now	VAS slider 0–10	VAS + ODI/NDI domain 1 (reused)
Q7	Pain pattern	Intermittent · Activity-related · Constant · Night pain	Clinical
Q8	Overall trend	Improving · Stable · Slowly worse · Rapidly worse	Clinical + registry
Q9	Radiating pain into arm/leg?	No · Occasional · Frequent · Constant	Neuro screen
Q10	Numbness or tingling?	None · Occasional · Frequent · Constant	Neuro screen
Q11	Weakness?	None · Mild · Moderate · Progressive	Neuro screen
Q12	Pain worsens with	Sitting · Standing · Walking · Forward bend · Backward bend · Lifting · Coughing (multi-select)	Mechanical pattern
Q13	Pain improves with	Rest · Walking · Changing position · Medicines · Heat/cold · Nothing	Mechanical pattern
Q14	Treatment tried so far	Medicines · Physio · Injection · Surgery · Alternative therapy · None (multi-select)	Treatment journey
Q15	Has treatment helped?	Significant improvement · Partial · No improvement · Worsened · Not yet tried	Treatment journey
Q16	Personal care (bathing/dressing) difficulty	None · Slight · Needs modification/help · Cannot do without help	Shared ODI+NDI domain
Q17	Lifting difficulty	None · Slight · Only light weights · Cannot lift	Shared ODI+NDI domain
Q18	Sleep disturbed by pain?	Not disturbed · Occasional · Frequent waking · Cannot sleep	Shared ODI+NDI domain

Q6, Q16, Q17 and Q18 are deliberately reused as the shared ODI/NDI domains (pain intensity, personal care, lifting, sleep) so the adaptive tail only needs to add the domains that differ between regions.

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Module 2 — Section C: Adaptive Functional Tail (Q19–Q24)

The system shows only **ONE** of the two sets below, based on the primary complaint region selected in Q1/Q2 — this keeps every patient's questionnaire at exactly 24 core questions while producing a fully valid, literature-comparable score for their region.

If primary complaint is **LOW BACK** → produces a fully valid ODI (Oswestry Disability Index)

#	Question	Response Options
Q19	Walking tolerance	Unlimited · 15–30 min · 5–15 min · <5 min
Q20	Sitting tolerance	>2 hrs · 30 min–2 hrs · <30 min · Cannot sit
Q21	Standing tolerance	>1 hr · 30–60 min · 10–30 min · <10 min
Q22	Social life affected?	Not affected · Slightly · Noticeably · Severely limited
Q23	Traveling affected?	Not affected · Slightly · Noticeably · Severely limited
Q24	Effect on intimate/personal relationships	Not affected · Slightly · Noticeably · Severely limited · Prefer not to answer

If primary complaint is **NECK** → produces a fully valid NDI (Neck Disability Index)

#	Question	Response Options
Q19	Reading affected?	Not affected · Slightly · Noticeably · Severely limited
Q20	Headache	None · Mild, infrequent · Moderate, frequent · Severe, constant
Q21	Concentration affected?	Not affected · Slightly · Noticeably · Severely limited
Q22	Work affected?	Not affected · Slightly · Noticeably · Unable to work
Q23	Recreation/leisure activities affected?	Not affected · Slightly · Noticeably · Severely limited
Q24	Driving affected?	Not affected · Slightly · Noticeably · Cannot drive

OUTCOME-TRACKING ANCHOR (counted separately from the 24)

Self-rated overall spine health today — 0 (worst) to 10 (completely healthy) slider. Used purely for 3/6/12-month outcome tracking, not for same-visit diagnosis.

CONDITIONAL ADD-ON — myelopathy screen (neck cases only)

Triggers only if Q1/Q2 indicate a neck-primary complaint AND Q10 (numbness) or Q11 (weakness) is answered "Frequent" or worse. Adds ~2 questions for this higher-risk subgroup only — everyone else does not see these.

Fine hand tasks (buttoning, writing, picking up small objects) — Normal · Slight clumsiness · Noticeable difficulty · Significant impairment

Balance/gait — Normal · Mild unsteadiness · Frequent imbalance · Needs support to walk

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Module 2 — Auto-Generated Scores (System Output, Not Shown to Patient)

Score / Tool	Source Questions	Applied When
VAS Pain Score	Q6	All patients
ODI — Oswestry Disability Index	Q6, Q16–Q18, Q19–Q24 (back tail)	Low-back primary complaint
NDI — Neck Disability Index	Q6, Q16–Q18, Q19–Q24 (neck tail)	Neck primary complaint
mJOA Myelopathy Screen	Red-flag screen + conditional add-on	Neck cases with frequent+ numbness/weakness
SEI-Score (triage)	Red-flag screen + Q6–Q11	All patients
Registry Baseline	Outcome-tracking anchor item	All patients — 3/6/12-month follow-up

SEI-Score Triage Pathway

Pathway	SEI-Score	Recommended Action
GREEN	0–25	Education + Rehabilitation
YELLOW	26–55	Routine specialist consultation
ORANGE	56–80	Priority specialist assessment
RED	>80 or any red flag	Urgent / same-day review

Labelling note for engineering: only label an output "ODI" or "NDI" when the full validated domain set for that instrument was actually used (i.e. the correct adaptive tail was shown). The SEI-Score is a proprietary triage tool and should always be labelled as such, not as a validated instrument.

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Module 3 — Document Upload

Simple upload screen, shown after the clinical questions. Can be completed before or shortly after booking — patients upload only what they have.

- ☐ Upload MRI — images or PDF report
- ☐ Upload X-rays
- ☐ Upload CT scan
- ☐ Upload Blood reports
- ☐ Upload Previous prescriptions
- ☐ Upload Surgery records
- ☐ Upload Referral letter
- ☐ Upload Insurance papers (*optional*)
- ☐ Upload Other documents

System automatically:

- ☐ Tags each upload by region (Lumbar / Cervical / Thoracic), date, and hospital
- ☐ Detects region from MRI metadata/report text where possible — patient is not asked to specify manually
- ☐ Maintains version history — a new MRI upload archives the previous one rather than replacing it

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System Output — Doctor Summary & Patient Confirmation

Once Registration, Clinical Pre-Consult Questions and Document Upload are all complete, two things happen automatically.

1. AI-GENERATED PRE-CONSULT SUMMARY (for the doctor, before consultation)

☐	Patient demographics, medical & surgical history, allergies, current medications, BMI
☐	Emergency/red-flag status (flagged prominently if positive)
☐	Symptom summary (region, duration, pattern, trend, mechanical behaviour)
☐	PROM scores: VAS, ODI or NDI (as applicable), mJOA screen (if triggered), SEI-Score & triage pathway
☐	Previous surgery details and previous MRI/imaging availability
☐	Uploaded reports, auto-tagged by region and date
☐	AI-generated probable diagnosis list and suggested investigations — for doctor validation, not final diagnosis

2. PATIENT CONFIRMATION MESSAGE (sent by SMS / WhatsApp / Email)

Thank you, **[Patient Name]**. Your pre-consult information has been received.

Your consultation is confirmed:

Date & Time: [Slot Date & Time]

Doctor: [Doctor Name], Dr. Vishal Kundnani Spine Unit

Teleconsultation Link: [Join Link]

Please join 5 minutes early. Keep your uploaded reports handy during the call.

If you develop any urgent symptoms before your appointment, contact us immediately at **[Spine Team Contact Number]**.

Delivery: sent automatically via the patient's preferred channel (SMS / WhatsApp / Email, as set during registration) immediately after Module 3 is submitted and a slot is allotted.

This specification is intended for the SEI product/engineering team to build against. Clinical wording of individual questions should undergo a final review by the treating clinical team before release.